

VHF/HF Radio Facility Inspection

A combined documentation, technical, and net-readiness checklist for examining a member's VHF and/or HF radio station offered as a U.S. Coast Guard Auxiliary operational facility.

Field aid, not the official record. This checklist is a working tool for flotilla examiners. It does not replace the current Auxiliary Manual (COMDTINST M16790.1), your District's Facility program SOPs, or the official Facility Examination form – confirm current form numbers, required signatures, and standards with your FSO-MS / SO-MS or DIRAUX before filing results.

OWNER / OPERATOR	CALLSIGN	FLOTILLA / DIVISION
FACILITY TYPE	BANDS OFFERED	INSPECTION DATE
Select	Select	mm/dd/yyyy
EXAMINER NAME / MEMBER #	PRIOR INSPECTION / RESULT	

A Documentation & Administrative Review

- ☐ Operator's Auxiliary membership and required Communications specialty training are current

- ☐ Facility offer / agreement for use is on file and matches equipment being inspected

- ☐ Equipment list (make, model, serial) matches what is physically installed

- ☐ Operator holds the license/authorization required for frequencies used
FCC restricted radiotelephone / amateur license (General+ for HF) / CG-issued authorization, as applicable

- ☐ Prior year's inspection report on file, deficiencies (if any) corrected

B VHF Equipment & Installation

- ☐ Transceiver powers up cleanly; make/model/serial recorded

- ☐ Transmit power within spec on high and low settings
Typ. ~25 W high / 1 W low, fixed or mobile marine VHF

- ☐ Ch. 16 and required working/weather channels programmed and receive clearly

- ☐ DSC function operational with correct MMSI; GPS input connected if equipped

- ☐ Antenna, coax, and connectors in good condition; connections weatherproofed

- ☐ SWR measured at feedpoint and within spec
≤ 2:1 acceptable, ≤ 1.5:1 preferred

- ☐ Antenna mount secure; coax routed with drip loop, no chafe points

C HF Equipment & Installation

- ☐ Transceiver powers up cleanly; make/model/serial recorded
- ☐ Frequency coverage includes assigned Auxiliary/CG and/or net working frequencies
- ☐ Transmit power matches operator's license class and equipment rating
- ☐ Antenna tuner present and achieves match across bands in use
- ☐ SWR acceptable on each band used for net operations
- ☐ Antenna type/height suits mission – NVIS for regional net coverage vs. higher/directional for long-haul
- ☐ Ground / counterpoise system adequate for antenna type

D Power Systems

- ☐ Primary power connection secure and properly fused
- ☐ Backup power source present and sized to run the station through an outage
Battery / generator / solar – confirm runtime against unit's expected net duration
- ☐ Battery voltage checked under transmit load
- ☐ Operator can demonstrate switch-over to backup power without assistance

E Grounding, Lightning & Safety

- ☐ Station ground rod / bond to structural ground present and intact
Verify per Appendix A procedure
- ☐ Coax surge suppressor and a means to disconnect/ground antenna feedlines when not in use
- ☐ Antenna placement keeps personnel clear of the RF exposure zone at rated power
- ☐ Cabling secured, no trip hazards; outdoor connections weatherproofed

F Operational & Net Readiness

- ☐ Voice check completed with NCS or another qualified station on the VHF working channel
- ☐ Voice check completed with NCS on the HF working frequency
Note propagation/time-of-day if signal was marginal
- ☐ Signal report obtained and logged
e.g. readability/strength "5x5" or RST
- ☐ Alternate/backup frequency known and tested
- ☐ Operator demonstrates net log procedure (paper or digital) and can record a check-in correctly
- ☐ Operator can describe emergency/priority traffic handling procedure

G Examiner Findings & Certification

PASS

CONDITIONAL

FAIL

Deficiencies noted / corrective action required

Describe any items marked unsatisfactory and the date by which they must be corrected.

EXAMINER SIGNATURE

DATE

mm/dd/yyyy



OWNER/OPERATOR SIGNATURE

NEXT INSPECTION DUE

mm/dd/yyyy



Grounding & Bonding Verification Procedure

Supports Section E. Use this procedure to actually verify each grounding item rather than eyeballing it â€” a station can look grounded and still have a broken bond, a corroded lug, or a lost RF ground that only shows up as a ruined SWR match or a shock hazard. Work through 1â€“5 in order; each depends on the one before it.

A.1 â€” concept **Single-Point Ground System**

1. Confirm the station has one common ground reference â€” a bus bar or heavy copper strap that every piece of equipment bonds to â€” rather than each item running to its own separate rod or cold-water pipe. Separate grounds at different earth potentials is what turns a lightning strike into a ground loop that arcs across the room.
2. Trace where AC safety ground (green wire), RF/station ground, and lightning ground meet. Per NEC practice they should bond together at one point near where service/antenna grounds enter the structure.

A.2 â€” antenna system **Mast/Tower and Feedline Ground**

1. Visually inspect the ground rod(s) at the antenna base for corrosion, a loose clamp, or exposed/nicked bonding wire.
2. Confirm the rod is bonded to the station ground with a short, direct run of at least #10 AWG copper wire or 3/4" copper strap â€” flat strap is preferred over round wire because it carries RF and transient surge current with less inductance.
3. Continuity test: meter between the antenna mount/mast and the ground rod clamp. Expect well under 1 Î©. A high or unstable reading means a corroded or loose bond â€” clean the contact point and retest.
4. If a ground resistance tester is available, measure the rod-to-earth resistance. Target â‰ˆ 25 Î© (NEC minimum for a single rod), ideally â‰ˆ 10 Î© for lightning protection. If it's high, bond in an additional rod at least 6 ft away.
5. Confirm a coax lightning arrestor/surge suppressor is inline on the feedline and its ground lug bonds to the same ground point â€” not a separate one.
6. Confirm there is a way to disconnect and ground the feedline when the station is not in use, particularly for portable/mobile setups during storm season.

Common finding: a arrestor installed but its ground lead run to a nearby fence post or downspout instead of the station ground â€” this defeats the single-point ground and should be corrected on the spot.

A.3 transceiver Chassis Ground

1. Locate the chassis ground lug on the rear panel (separate from the DC power or coax connections).
2. Confirm a dedicated ground strap runs from that lug directly to the station ground bus – the AC power cord's safety ground is not a substitute for this bond.
3. Keep the strap as short as physically possible, and well under a quarter-wavelength of the lowest operating frequency, so it can't become a resonant radiator and re-introduce RF into the shack. Flat braid is preferred over round wire for the same low-inductance reason as A.2.
4. Check the mounting point is bare metal – no paint or anodizing under the lug – and the hardware (including a star washer) is tight.
5. Continuity test chassis-to-ground-bus: expect well under 1 Ω .

A.4 tuner Antenna Tuner Ground

1. Locate the tuner's dedicated ground terminal – separate from the coax shield connection.
2. Confirm it bonds directly to the station ground bus with a short, heavy strap. Many external tuners rely on this connection as part of the RF counterpoise, especially when matching a random-wire or end-fed antenna – it is not just a safety ground.
3. If the operator reports the tuner can no longer find a match on a band it used to tune cleanly, check this bond first before suspecting the antenna itself – a lost or corroded tuner ground is the most common cause.
4. For end-fed or random-wire antennas, confirm the counterpoise wire(s) specified by the tuner's manual are present, undamaged, and connected – the tuner ground alone is rarely sufficient without them.

A.5 associated equipment Power Supplies, Amplifiers, Switches & Interfaces

1. Bond the metal chassis of every associated item – power supply, amplifier, coax switch, digital-mode/logging interface – to the same ground bus used in A.3. Never give one item its own independent ground rod; two ground points at different potentials is how ground loops and hum/RFI get introduced.
2. For a coax switch, verify the body is grounded and check continuity through each position, not just the one currently in use.
3. For a computer interface used for digital modes or NetLog-style logging, note whether it is opto-isolated. A non-isolated interface is a common source of RF feedback and ground-loop hum and may need an isolation transformer or ferrite choking added.

A.6 safety Precautions While Testing

1. Never test or work on the grounding system during a thunderstorm or when lightning is possible in the area.

2. De-energize the station before disconnecting or disturbing any bonding strap.
3. Use insulated gloves and insulated-handle tools when probing near an energized station.
4. After any known lightning event near the station, re-inspect and re-test the entire ground system in full before returning the station to service.

A.7 “ record Verification Log

POINT TESTED	READING	RESULT	NOTES
Antenna mast ‘ ground rod	10		✓
Ground rod ‘ earth	10		✓
Transceiver chassis ‘ bus	10		✓
Tuner ground ‘ bus	10		✓
Other equipment ‘ bus	10		✓

Suggested cadence: re-examine at least annually, and after any change of equipment, antenna, or installation location. Retain a copy with the flotilla's facility records and forward per your District's process.